

Anil Kumar

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CAREER OBJECTIVE

M.Tech Data Science student at Amrita Vishwa Vidyapeetham with hands-on experience across Deep Learning, Computer Vision, NLP, and Time-Series Analysis. Designs end-to-end PyTorch pipelines from dataset construction to model training, evaluation, and deployment. Seeking Data Science / Machine Learning / AI / Computer Vision / GenAI roles.

TECHNICAL SKILLS

Programming & Libraries

Python, NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Seaborn, Plotly, Hugging Face Transformers, OpenCV

Time-Series

Filtering, Peak Detection, Segmentation, ARIMA, SARIMA

Deep Learning

PyTorch, TensorFlow, Keras, CNN, LSTM, GRU, BiLSTM, ViT, Swin, ConvNeXt, EfficientNet, MobileNet, CLIP, Transformer Decoder, CBAM, AMP, CUDA

Machine Learning

Regression, Classification, Feature Engineering, PCA, Transfer Learning, Fine-Tuning, Hyperparameter Tuning, Focal Loss, TTA, SHAP Explainability

Statistical Analysis & EDA

Exploratory Data Analysis (EDA), Hypothesis Testing, Correlation Analysis, Distribution Analysis, Outlier Detection, Statistical Modeling

NLP

Transformers, Hugging Face, BERT, spaCy, NLTK, Tokenization, Text Preprocessing, Named Entity Recognition (NER), Text Classification

Computer Vision

Image Processing, Feature Extraction, Object Detection, Fine-Grained Classification, Data Augmentation

Databases, Backend & Cloud

Docker, Git, GitHub, REST APIs, AWS, Jupyter Notebook

EDUCATION

Amrita Vishwa Vidyapeetham, Bengaluru

M.Tech in Data Science

2025 – Present

CGPA: 8.16

Vel Tech R&D Institute of Science and Technology, Chennai

B.Tech in Computer Science

2017 – 2021

CGPA: 7.44

Sri Chaitanya Junior College, Tirupathi

Class XII (Intermediate)

2017

Percentage: 94.4%

PROJECTS

Kannada Image Captioning: Dataset Construction and Benchmark Evaluation of CNN-RNN and Transformer Architectures 2026

Technologies: Python, PyTorch, IndicTrans2-1B, CLIP, ViT, Swin, ConvNeXt, IndicBART (BPE), BERTScore, sacrebleu

- Constructed a **250K-caption Kannada corpus** by translating 50K MS COCO images via IndicTrans2-1B.
- Implemented and benchmarked **6 encoder-decoder architectures** spanning CNN-RNN with Bahdanau attention (EfficientNet, MobileNet, Swin with LSTM/GRU) and transformer decoders on ConvNeXt, ViT, and CLIP ViT-B/16.
- Best model ConvNeXtTransformer reached **BLEU-4 = 67.95, chrF = 42.95**.
- Validated end-to-end semantics via Kannada→English back-translation with BERTScore on 10K test images: **BERTScore F1 = 84.02**.

AgroVision-7: Lightweight Deep Learning for Fine-Grained Rice Leaf Disease Recognition 2026

Technologies: Python, PyTorch, ViT, Swin, ConvNeXt, Optuna, Focal Loss, TTA, Ensemble Learning, Grad-CAM

- Built a 7-class fine-grained rice leaf disease classifier on 14,241 images, benchmarking 4 architecture families under a single imbalance-aware protocol; final ensemble of ViT-B/16, Swin-T, and ConvNeXt-Tiny achieved **97.48% accuracy and 0.9750 macro-F1**, with every per-class accuracy ≥ 0.96 .
- Engineered an imbalance-mitigation pipeline using **WeightedRandomSampler** and class-weighted Focal Loss, lifting ResNet-50 by **+17.23 accuracy points** over the baseline; trained with two-stage transfer learning, AMP, gradient clipping, and label smoothing.
- Ran a 15-trial Optuna (TPE) hyperparameter search; applied 5-view deterministic TTA, accuracy-weighted softmax ensembling, and Grad-CAM explainability.

Mental Health Risk and Treatment Prediction using Machine Learning with Agentic AI Integration 2025

Technologies: Python, NumPy, Pandas, SciPy, scikit-learn, XGBoost, LightGBM, SHAP, EDA, Statistical Analysis, Stacking Ensemble, Grid Search, Bayesian Optimization

- Designed a **reproducible end-to-end ML pipeline** on a clinical-style mental-health survey dataset; performed structured

- EDA, statistical analysis, distribution checks, and missing-value diagnostics, then benchmarked 11 classifiers .
- Ran controlled experiments comparing **Grid Search vs. Bayesian Optimization (Optuna)** across five tree-based models; XGBoost improved by **+1.3 accuracy and +1.5 ROC-AUC points** post-tuning, validating Bayesian Optimization as the more sample-efficient strategy on this dataset.
 - Achieved **95.22% accuracy and 0.9939 ROC-AUC** with a Stacking Ensemble of optimized base learners; documented assumptions, limitations, and feature-stability findings across cross-validation splits.
 - Applied **SHAP-based explainability** and tree-based feature-importance consensus (96% cross-model agreement) to identify clinically meaningful risk indicators — aligning model outputs with interpretable, decision-support criteria suitable for healthcare contexts.

PUBLICATIONS & CONFERENCES

Wildfire Occurrence Analysis Using Meteorological Data: A Comparative Study of Statistical Methods and Learning Models

9th Intl. Conference on Inventive Computation Tech. (ICICT 2026)

April 2026

CERTIFICATIONS

AWS Academy Graduate – Cloud Foundations – Amazon Web Services

Feb 2026